LGM-GEN: a mode-space generator for type-specific mutual excitation in a marked temporal point process

Mathematical specification

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Abstract

We specify LGM-GEN, a marked temporal point process for limit-order-book event flow. The model keeps a scalar, *type-blind* linear Hawkes ground intensity whose branching ratio and mean rate are pinned in closed form, and places all mark-to-mark mutual excitation inside the mark simplex, carried by R latent modes with a dense generator G parameterised in its eigenbasis. Because the mark distribution is normalised at every t, the total intensity is algebraically independent of every generator parameter. Three properties follow exactly rather than approximately: the counting process retains the type-blind law (so the branching ratio, the rate pin and the Fano factor are invariant); the log-likelihood remains additively separable (so the transplant property holds); and the lag-integrated mark-to-mark transition matrix has a closed form, $M = \mathrm{Re}\big(U\,\mathrm{diag}(c)(-\Lambda_G)^{-1}V^{\top}\big)$, which is directly comparable to an independently measured binned-count regression estimate.

1 Setup and notation

Let $(t_i, x_i)_{i\geq 1}$ be an event stream with $0 < t_1 < t_2 < \cdots$ and marks $x_i \in \{0,1\}^K$, multi-hot indicators over K channels; a single event may activate several channels simultaneously, so marks are *sets*. Write $\mathcal{H}_t = \sigma\{(t_i, x_i) : t_i < t\}$ and $|x| = \sum_k x_k$. In the deployed configuration $K = 62$, indexing $\{\mathsf{LO}, \mathsf{CO}\} \times \{\text{bid}, \text{ask}\} \times \{L_1, \ldots, L_{10}\}$, $\mathsf{MO} \times \{\text{bid}, \text{ask}\}$, and $\mathsf{IS} \times \{\text{bid}, \text{ask}\} \times \{L_1, \ldots, L_{10}\}$.

The model factorises the channel intensity into a scalar ground and a distribution on the simplex,

$$\lambda_k(t) \;=\; \Lambda(t)\, p(k \mid \mathcal{H}_t), \qquad \sum_{k=1}^{K} p(k \mid \mathcal{H}_t) = 1 \quad \text{for all } t. \tag{1}$$

Every structural result below follows from the normalisation in (1).

2 Ground intensity: scalar, type-blind, pinned

Fix M timescales and define decayed event counts

$$S^m(t) = \sum_{t_i < t} e^{-\beta_m (t - t_i)}, \qquad \beta_m = \mathrm{softplus}(\theta^{\delta}_m) + \delta_{\min}, \qquad m = 1, \ldots, M, \tag{2}$$

with $\delta_{\min} = 5 \times 10^{-3}$. The ground is linear in these accumulators,

$$\boxed{\Lambda(t) = \mu_0 + \sum_{m=1}^{M} a_m S^m(t), \qquad a_m = \mathrm{softplus}(\alpha_m) \geq 0.} \tag{3}$$

Crucially S^m counts events *without reference to their marks.* The ground is therefore type-blind by construction, and two identities hold exactly rather than asymptotically:

$$\text{branching ratio} \qquad n = \sum_{m=1}^{M} \frac{a_m}{\beta_m}, \tag{4}$$

$$\text{rate pin} \qquad \mu_0 = R_{\text{target}}\,(1-n) \implies \mathbb{E}[\Lambda] = \frac{\mu_0}{1-n} = R_{\text{target}}. \tag{5}$$

Equation (4) is gauge-free: it depends on no normalisation convention and is computable in closed form from the parameters alone. Deployed values for the SOL instrument are $M = 6$, $n = 0.9900$, $R_{\text{target}} = 22.2569$ events s^{-1}, and $\beta = (100,\ 18.206,\ 3.3145,\ 0.6034,\ 0.1099,\ 0.02)\,\mathrm{s}^{-1}$.

3 Backbone

A stacked diagonal state-space encoder (S2P2) produces the mark-head input. With L layers of H modes and channel embeddings $E \in \mathbb{R}^{K\times d_E}$,

$$e_i = \frac{1}{|x_i|}\sum_{k=1}^{K} x_{i,k}E_k, \tag{6}$$

$$\Delta^{(\ell)}(u) = \delta^{(\ell)} \odot \operatorname{clamp}\big(\operatorname{softplus}(W^{(\ell)}_{\text{dyn}}u),\ 0.05,\ 20\big), \tag{7}$$

$$x^{(\ell)}(t) = \bar{a}\odot x^{(\ell)}(t_i^{+}) + (\mathbf{1}-\bar{a})\odot W^{(\ell)}_{\text{in}}u^{(\ell-1)}, \qquad \bar{a} = e^{-\Delta^{(\ell)}(t-t_i)}, \tag{8}$$

and $h(t) \in \mathbb{R}^H$ is the layer-normalised top-layer output. Deployed: $L = 2$, $H = 64$, $d_E = 64$. All decay rates are real and positive, so the backbone spans mixtures of real exponentials; it admits no oscillatory kernel.

4 Mode-space generator

This is the component that encodes mutual excitation between marks.

4.1 Parameterisation

Let R be the number of latent impact modes, with parameters $\theta^\delta, \omega \in \mathbb{R}^R$, an emission map $V \in \mathbb{R}^{K\times R}$, and a reception map $U = U^{\text{re}} + iU^{\text{im}}$, $U^{\text{re}}, U^{\text{im}} \in \mathbb{R}^{K\times R}$. Set

$$\delta_r = \operatorname{softplus}(\theta^\delta_r) + \delta_{\min} > 0, \qquad \lambda_r = -\delta_r + i\omega_r, \qquad \Lambda_G = \operatorname{diag}(\lambda_1,\dots,\lambda_R). \tag{9}$$

The object represented is a *dense* generator G on mode space, written in its eigenbasis,

$$G = P\,\Lambda_G\,P^{-1}. \tag{10}$$

Only the spectrum Λ_G is identifiable: for any invertible P the substitution $(V, U) \mapsto (VP^{-\top}, UP)$ leaves every observable unchanged. We therefore fold P into V and U and never materialise G. The model *computes* as a complex-diagonal state-space model — preserving an $O(\log N)$ associative scan — while *representing* a dense generator.

4.2 Dynamics

The mode state $z(t) \in \mathbb{C}^R$ evolves by the generator between events and jumps at events:

$$\boxed{\dot{z}(t) = \Lambda_G \, z(t), \qquad z(t_i^+) = z(t_i^-) + V^\top x_i.} \tag{11}$$

Equivalently, in closed form and in the recursion actually evaluated,

$$z(t) = \sum_{t_i<t} \big(V^\top x_i\big) \, e^{\lambda(t-t_i)}, \tag{12}$$

$$z(t_i^-) = e^{\lambda\,\Delta t_i}\Big(z(t_{i-1}^-) + V^\top x_{i-1}\Big), \qquad \Delta t_i = t_i - t_{i-1}, \;\; t_0 := 0. \tag{13}$$

Emission is *sum*-pooled, $V^\top x_i = \sum_k x_{i,k} V_{k\cdot}$, so an event activating j channels emits j rows; set cardinality, which the backbone's mean pooling discards, is retained here.

Since $\delta_r > 0$ we have $|e^{\lambda_r \Delta t}| = e^{-\delta_r \Delta t} < 1$ for $\Delta t > 0$, so the recursion (13) is unconditionally stable and requires no spectral-radius certificate.

4.3 Mark head

Let $s(t) = \mathrm{MLP}(h(t)) \in \mathbb{R}^K$ be the backbone mark score. The generator contributes an additive logit residual, with per-mode conditioning $c_r = \delta_r / R_{\text{target}}$ chosen so that $c_r \, \mathbb{E}|z_r| = O(1)$ uniformly over timescales spanning four decades:

$$\Delta s(t) = \mathrm{Re}\big(U\,(c \odot z(t))\big) = \sum_{r=1}^{R} c_r \Big(U_{\cdot r}^{\text{re}} \,\mathrm{Re}\, z_r(t) - U_{\cdot r}^{\text{im}} \,\mathrm{Im}\, z_r(t)\Big), \tag{14}$$

$$p(k \mid \mathcal{H}_t) = \frac{\exp\big(s_k(t) + \Delta s_k(t)\big)}{\sum_{j=1}^{K} \exp\big(s_j(t) + \Delta s_j(t)\big)}. \tag{15}$$

5 The per-type intensity, fully expanded

Substituting (12) into (14) and exchanging the order of summation shows that the generator residual is itself a Hawkes sum, but in logit space:

$$\Delta s_k(t) = \sum_{t_i<t} \sum_{j=1}^{K} x_{i,j} \, \varphi_{kj}(t - t_i), \tag{16}$$

with the pair kernel φ of (20). The inner sum over j absorbs set-valued marks: every channel active in event i contributes its own row. Combining (1), (3), (15) and (16),

$$\boxed{\lambda_k(t) = \underbrace{\Big[R_{\text{target}}(1-n) + \sum_{m=1}^{M} a_m \sum_{t_i<t} e^{-\beta_m(t-t_i)}\Big]}_{\text{scalar, type-blind}} \cdot \frac{\exp\Big(s_k(t) + \sum_{t_i<t}\sum_j x_{i,j}\varphi_{kj}(t-t_i)\Big)}{\sum_{l=1}^{K} \exp\Big(s_l(t) + \sum_{t_i<t}\sum_j x_{i,j}\varphi_{lj}(t-t_i)\Big)},} \tag{17}$$

or equivalently

$$\log \lambda_k(t) = \log\Big[R_{\text{target}}(1-n) + \sum_m a_m S^m(t)\Big] + s_k(t) + \sum_{t_i<t}\sum_j x_{i,j}\varphi_{kj}(t-t_i) - \log \sum_l e^{s_l + \Delta s_l}. \tag{18}$$

Remark 1 (Reading the form). The model is a linear Hawkes process multiplied by a softmax of a linear Hawkes process. Both valves are exponential-kernel sums over the past; they differ only in that the ground's kernel is untyped and enters the *rate* additively, while the generator's is fully typed and enters the *logit* additively. The log-normalizer is the sole coupling between the K logits, and it is exactly what enforces $\sum_k \lambda_k(t) = \Lambda(t)$ for any φ — so Propositions 1–2 are visible directly in (17).

Remark 2 (Contrast with multivariate Hawkes). A conventional multivariate Hawkes process would set $\lambda_k(t) = \mu_k + \sum_{t_i<t}\sum_j x_{i,j}\varphi^H_{kj}(t-t_i)$, additive in the rate. There φ^H *creates* events, so $\int \varphi^H$ is a branching matrix whose spectral radius governs stability. In (17), φ only *reallocates* a rate fixed elsewhere; consequently the induced effective branching matrix has all column sums equal to n, and $\|\varphi\|$ carries no stability meaning for the counting process.

Remark 3 (The one non-Hawkes term). $s_k(t) = \mathrm{MLP}(h(t))_k$ is nonlinear in the history and is the only term preventing (17) from being a closed-form Hawkes process. Freezing $s \equiv \mathrm{const}$ reduces the mark process to an exactly analysable multivariate Hawkes process in logits.

6 Likelihood and structural properties

For a stream observed on $[0,T]$, equation (1) gives

$$\log\mathcal{L} = \underbrace{\sum_i \log\Lambda(t_i) - \int_0^T \Lambda(u)\,du}_{\ell_{\text{ground}}} \;+\; \underbrace{\sum_i \log p(k_i \mid \mathcal{H}_{t_i})}_{\ell_{\text{mark}}}. \tag{19}$$

Write $\Theta_{\text{gen}} = (V, U^{\text{re}}, U^{\text{im}}, \theta^\delta, \omega)$ for the generator parameters and $\Theta_{\text{gr}} = (\alpha, \theta^\delta_{\text{ground}})$ for the ground.

Proposition 1 (Intensity independence)**.** *$\partial\Lambda(t)/\partial\Theta_{\text{gen}} \equiv 0$ for all t.*

Proof. By (3), Λ is a function of (μ_0, a, β) and the accumulators S^m. By construction S^m depends on the event *times* only. Θ_{gen} enters the model solely through (14) and hence through (15), which affects p but not Λ. □

Proposition 2 (Count autonomy)**.** *The law of the counting process $N(t) = \#\{i : t_i \le t\}$ is independent of Θ_{gen}. Consequently n, the pin $\mathbb{E}[\Lambda] = R_{\text{target}}$, and the Fano factor $F(\Delta) = \mathrm{Var}\,N_\Delta/\mathbb{E}N_\Delta$ are invariant.*

Proof. N is a point process with conditional intensity Λ, which by Proposition 1 does not depend on Θ_{gen}; and Λ depends on the past only through event times, so the time-marginal recursion closes on itself. Hence the finite-dimensional distributions of N are unchanged. The three quantities are functionals of that law: n and μ_0 via (4)–(5), and F via the Cox representation $F(\Delta) = 1 + \mathrm{Var}(\Lambda_\Delta)/\mathbb{E}[\Lambda_\Delta]$. □

Proposition 3 (Separability and transplant)**.** *ℓ_{ground} depends only on Θ_{gr} and ℓ_{mark} only on $(\Theta_{\text{gen}}, \Theta_{\text{backbone}})$. Hence $\log\mathcal{L}$ is additively separable, and replacing Θ_{gr} by any other ground parameterisation leaves the maximiser of ℓ_{mark} unchanged: a mark head fitted under one ground is exactly optimal under any other.*

Proof. Immediate from (19) and Proposition 1: no parameter appears in both blocks, so $\nabla_{\Theta_{\text{gen}}}\log\mathcal{L} = \nabla_{\Theta_{\text{gen}}}\ell_{\text{mark}}$ and $\nabla_{\Theta_{\text{gr}}}\log\mathcal{L} = \nabla_{\Theta_{\text{gr}}}\ell_{\text{ground}}$. □

Remark 4 (Why the structure sits in the composition). An alternative design multiplies the ground kick by a per-type weight w_k, giving $n_{\text{eff}} = n \cdot w_k$ locally. Near criticality this has almost no admissible range: at $n = 0.99$, stability of every type requires $\max_k w_k < 1/n \approx 1.0101$, a 1% budget, whereas regression-identified weights span 1.02–6.28 across instruments. Empirically that design inflates the Fano factor by 89%–297%. Routing inside the simplex has no such budget: by Proposition 2 its cost is exactly zero.

7 Induced mark-to-mark transition matrix

Because the readout is linear in z and z is linear in past emissions, the model induces an explicit kernel on mark space. The logit-space impulse response of channel k to a channel-j event τ seconds earlier is

$$\varphi_{kj}(\tau) = \mathrm{Re}\Big(\sum_{r=1}^{R} U_{kr}\, c_r\, e^{\lambda_r \tau}\, V_{jr}\Big) = \sum_{r=1}^{R} c_r\, e^{-\delta_r \tau}\Big(U^{\text{re}}_{kr} \cos\omega_r\tau - U^{\text{im}}_{kr} \sin\omega_r\tau\Big) V_{jr}. \tag{20}$$

Since $\mathrm{Re}\,\lambda_r < 0$, the lag integral converges and $\int_0^\infty e^{\Lambda_G \tau} d\tau = (-\Lambda_G)^{-1}$, giving a closed form for the lag-integrated transition matrix:

$$\boxed{M = \int_0^\infty \varphi(\tau)\, d\tau = \mathrm{Re}\Big(U\ \mathrm{diag}(c)\,(-\Lambda_G)^{-1} V^\top\Big), \qquad M_{kj} = \mathrm{Re} \sum_{r=1}^{R} \frac{U_{kr}\, c_r\, V_{jr}}{\delta_r - i\omega_r}.} \tag{21}$$

Remark 5 (Units). M is a shift in log-odds, i.e. a *composition* operator. It is not a branching matrix and its spectral radius carries no stability meaning; it is comparable in *pattern*, but not in scale, to an offspring matrix estimated by binned-count regression.

Remark 6 (What a dense G buys). A real diagonal generator with freely signed U, V already spans mixtures of real exponentials, which includes delayed peaks of the form $c(e^{-a\tau} - e^{-b\tau})$. The strict additions from $\omega \neq 0$ are damped oscillation, $e^{-\delta\tau} \cos\omega\tau$, unreachable by a real-positive diagonal model; and, more importantly, parameter sharing: one $R \times R$ spectrum governs all K^2 pairs through (20), so the kernel is constrainable and comparable to measurement, whereas an unstructured backbone learns each pair's mixture independently.

8 Parameters and fitted values

Block	Count	Status
$\theta^\delta,\ \omega$	$2R = 64$	trained
V	$KR = 1{,}984$	trained
$U^{\text{re}},\ U^{\text{im}}$	$2KR = 3{,}968$	trained
Generator total	**6,016**	trained
Backbone, mark MLP, ground	57,033	frozen

Table 1: Deployed configuration $K = 62$, $R = 32$, $H = 64$, $L = 2$, $M = 6$. The generator is fitted as a logit residual with U initialised to zero, so the model reproduces its donor exactly at initialisation and every reported change is attributable to the generator.

Fitted on the SOL instrument (seed 3): mode timescales $1/\delta_r \in [0.048, 113]$ s; 16 of 32 modes oscillatory with periods in $[0.49, 675]$ s; $\max_{kj} |M_{kj}| = 0.0449$.

9 Empirical verification of the invariants

Quantity	baseline	LGM-GEN	magnitude-based variant
Branching ratio n	0.990000	0.990000	0.990000
Time NLL / event	-3.0349947	-3.0349947	-3.019058
Fano Δ vs. baseline, 1–50 s	—	+1.0% to +3.2%	+88.7% to +297.4%
Mark perplexity	23.9575	23.8665	23.9575
$p(\mathsf{MO})$ rollout / empirical	51.7%	74.0%	36.8%

Table 2: SOL, seed 3; Fano averaged over three rollout seeds. Time NLL is bit-identical over 1.98×10^6 held-out events, confirming Proposition 1 numerically. The residual Fano deviation is sampler-stream divergence, not a change of law: altered mark draws consume different pseudo-random state and hence shift subsequent time draws. For scale, the baseline's own across-seed spread is 14.8%–49.5% over the same scales. The magnitude-based variant places the same measured information in the ground kick rather than the composition.

10 Summary

The design principle is a strict division of labour. The ground determines *how much* happens and is type-blind, which buys exact control of the branching ratio, the mean rate and the dispersion. The generator determines *what* happens and is type-specific, expressive enough to carry damped oscillation and shared structure across all K^2 mark pairs, and — by Propositions 1–3 — free of any cost to the counting process. Mutual excitation is thereby made explicit and measurable through (21) without spending any of the stability budget that criticality leaves.

A Step-by-step derivation

Each step below is a single substitution, so the route from the primitives to the boxed intensity (17) can be checked line by line.

Step 1: the factorisation. The model asserts that the K-dimensional intensity splits into a scalar and a distribution,

$$\lambda_k(t) = \Lambda(t)\, p(k \mid \mathcal{H}_t), \qquad \sum_k p(k \mid \mathcal{H}_t) = 1.$$

Two objects require expansion: Λ and p.

Step 2: ground accumulators. Each past event contributes an exponential at M timescales, *irrespective of its mark*,

$$S^m(t) = \sum_{t_i < t} e^{-\beta_m (t - t_i)}, \qquad \beta_m = \mathrm{softplus}(\theta^\delta_m) + \delta_{\min}.$$

Step 3: the pin. With $\Lambda(t) = \mu_0 + \sum_m a_m S^m(t)$, take expectations in stationarity. Each accumulator satisfies $\mathbb{E}[S^m] = \mathbb{E}[\Lambda]/\beta_m$, hence

$$\mathbb{E}[\Lambda] = \mu_0 + \mathbb{E}[\Lambda] \sum_m \frac{a_m}{\beta_m} = \mu_0 + n\, \mathbb{E}[\Lambda] \qquad \Longrightarrow \qquad \mathbb{E}[\Lambda] = \frac{\mu_0}{1-n}.$$

Choosing $\mu_0 = R_{\text{target}}(1-n)$ therefore forces $\mathbb{E}[\Lambda] = R_{\text{target}}$, and

$$\Lambda(t) = R_{\text{target}}(1-n) + \sum_{m=1}^{M} a_m \sum_{t_i<t} e^{-\beta_m(t-t_i)}, \qquad n = \sum_m \frac{a_m}{\beta_m}.$$

No mark index appears: the first factor is complete and type-blind.

Step 4: generator state. Emission sums the rows of V selected by the mark set, $v_r(x_i) = (V^\top x_i)_r = \sum_j x_{i,j} V_{jr}$, and the state accumulates emissions with complex decay $\lambda_r = -\delta_r + i\omega_r$:

$$z_r(t) = \sum_{t_i<t} v_r(x_i)\, e^{\lambda_r(t-t_i)} \in \mathbb{C}.$$

Step 5: real and imaginary parts. Writing $\tau_i = t - t_i$ and $e^{\lambda_r \tau} = e^{-\delta_r \tau}(\cos\omega_r\tau + i \sin\omega_r\tau)$,

$$\operatorname{Re} z_r(t) = \sum_i v_r(x_i) e^{-\delta_r\tau_i} \cos\omega_r\tau_i, \qquad \operatorname{Im} z_r(t) = \sum_i v_r(x_i) e^{-\delta_r\tau_i} \sin\omega_r\tau_i.$$

Step 6: readout.

$$\Delta s_k(t) = \sum_{r=1}^{R} c_r \Big(U^{\text{re}}_{kr} \operatorname{Re} z_r(t) - U^{\text{im}}_{kr} \operatorname{Im} z_r(t) \Big), \qquad c_r = \delta_r / R_{\text{target}},$$

which is $\operatorname{Re}\big(U(c \odot z)\big)_k$ with $U = U^{\text{re}} + iU^{\text{im}}$, since $\operatorname{Re}[(a+bi)(x+yi)] = ax - by$.

Step 7: exchange the summations. Substituting Step 5 into Step 6 gives

$$\Delta s_k(t) = \sum_r c_r \Big[U^{\text{re}}_{kr} \sum_i v_r(x_i) e^{-\delta_r\tau_i} \cos\omega_r\tau_i - U^{\text{im}}_{kr} \sum_i v_r(x_i) e^{-\delta_r\tau_i} \sin\omega_r\tau_i \Big].$$

The sums are finite and $|e^{\lambda_r\tau}| \le 1$, so reordering is unconditionally valid. Pulling $\sum_i$ outward and expanding $v_r = \sum_j x_{i,j} V_{jr}$,

$$\Delta s_k(t) = \sum_{t_i<t} \sum_j x_{i,j} \underbrace{\Big[\sum_r c_r V_{jr} e^{-\delta_r\tau_i} \big(U^{\text{re}}_{kr} \cos\omega_r\tau_i - U^{\text{im}}_{kr} \sin\omega_r\tau_i \big) \Big]}_{\varphi_{kj}(\tau_i)}.$$

The bracket depends only on (k, j, τ); it is the pair kernel of (20), and the result is (16) — a Hawkes sum in logit space.

Step 8: mark distribution. $p(k \mid \mathcal{H}_t) = \exp(s_k + \Delta s_k) \big/ \sum_l \exp(s_l + \Delta s_l)$ with $s(t) = \mathrm{MLP}(h(t))$.

Step 9: assemble. Multiplying Step 3 by Step 8 yields the boxed expression (17).

Step 10: lag integral in real form. Since $\operatorname{Re}\lambda_r < 0$ we have $\int_0^\infty e^{\lambda_r \tau} d\tau = (\delta_r - i\omega_r)^{-1}$; rationalising $(\delta - i\omega)^{-1} = (\delta + i\omega)/(\delta^2 + \omega^2)$ turns (21) into purely real arithmetic,

$$M_{kj} = \sum_{r=1}^{R} c_r \, V_{jr} \, \frac{U^{\mathrm{re}}_{kr}\delta_r - U^{\mathrm{im}}_{kr}\omega_r}{\delta_r^2 + \omega_r^2}.$$

Step 11: the evaluated recursion. With $\Delta t_i = t_i - t_{i-1}$, $\bar{a}_r = e^{-\delta_r \Delta t_i}$ and $\vartheta_r = \omega_r \Delta t_i$, the state at t_i^- is obtained from that at t_{i-1}^- by

$$\begin{aligned}
\operatorname{Re} z_r &\leftarrow \bar{a}_r \big[(\operatorname{Re} z_r + v_r) \cos\vartheta_r - \operatorname{Im} z_r \sin\vartheta_r \big], \\
\operatorname{Im} z_r &\leftarrow \bar{a}_r \big[(\operatorname{Re} z_r + v_r) \sin\vartheta_r + \operatorname{Im} z_r \cos\vartheta_r \big],
\end{aligned}$$

i.e. $O(R)$ per event with no history retained; this is what the associative doubling scan evaluates in $O(\log N)$ depth.

Consistency checks

- **Normalisation.** $\sum_k \lambda_k(t) = \Lambda(t) \sum_k p_k = \Lambda(t)$ for any φ whatsoever, so Propositions 1 and 2 are readable directly off (17).
- **$U \to 0$.** Then $\varphi \equiv 0$, $\Delta s \equiv 0$, and the model collapses to its donor exactly — which is why zero-initialisation reproduces the donor bit-for-bit.
- **$\omega \to 0$.** Then $\varphi_{kj}(\tau) = \sum_r c_r V_{jr} U^{\mathrm{re}}_{kr} e^{-\delta_r \tau}$ and $M_{kj} = \sum_r c_r V_{jr} U^{\mathrm{re}}_{kr}/\delta_r$: a pure real-exponential mixture. The ω terms are exactly what a real-diagonal backbone cannot express.
- **Dimensions.** M carries units of logit·seconds and R_{target} of events/second, so $R_{\mathrm{target}} M$ is dimensionless — which is why R_{target}, and not n, is the conversion factor between the transition matrix and offspring counts in the effective branching matrix $A_{kj} = n\bar{p}_k + R_{\mathrm{target}} \bar{p}_k (M_{kj} - \langle M_{\cdot j} \rangle_{\bar{p}})$.